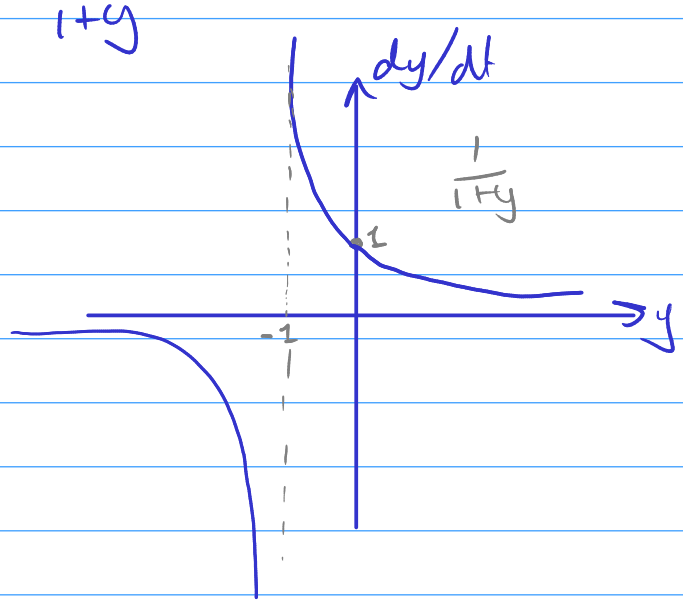
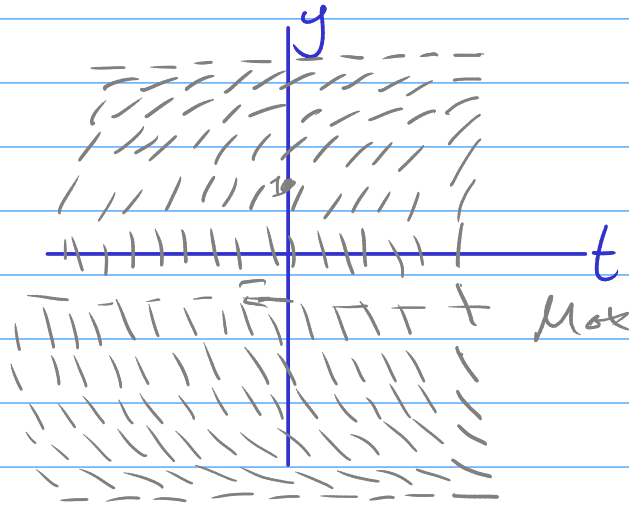
Worksheet C

Q1 Direction fields.

a) $\frac{dy}{dt} = \frac{1}{1+y}$



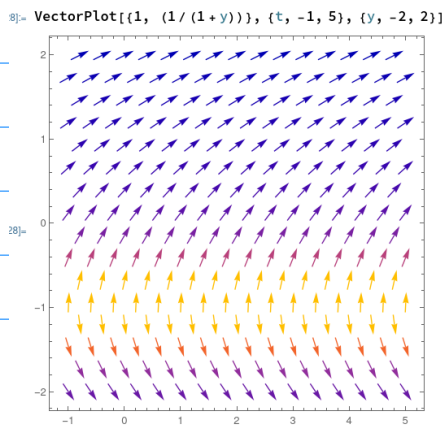
Direction plot



Max gradient @ $y=-1$

Solving $\int dy(1+y) = \int dt$

$y + y^2 + c = t$



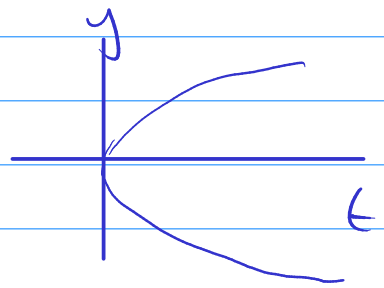
Complete the square

$$y^2 + y + \frac{1}{4} = c + t + \frac{1}{4}$$

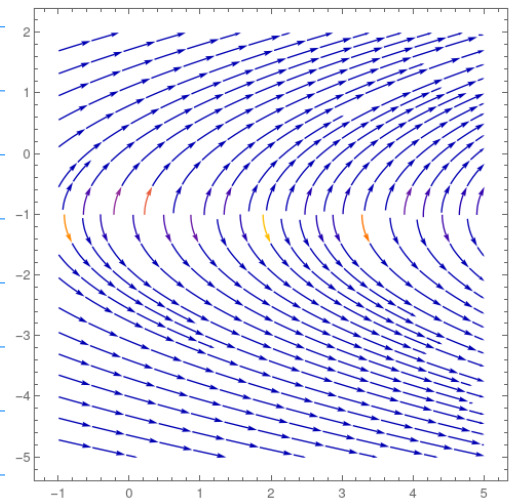
$$= \left(y + \frac{1}{2}\right)^2 = c + t + 1/4$$

$$y = \sqrt{t + c + 1/4} - \frac{1}{2}$$

$y \propto \sqrt{t}$ Which has shape

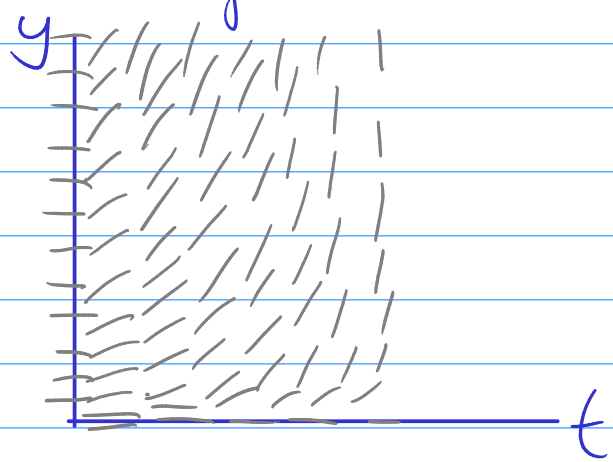


StreamPlot[{1, (1/(1+y))}, {t, -1, 5}, {y, -5, 2}]



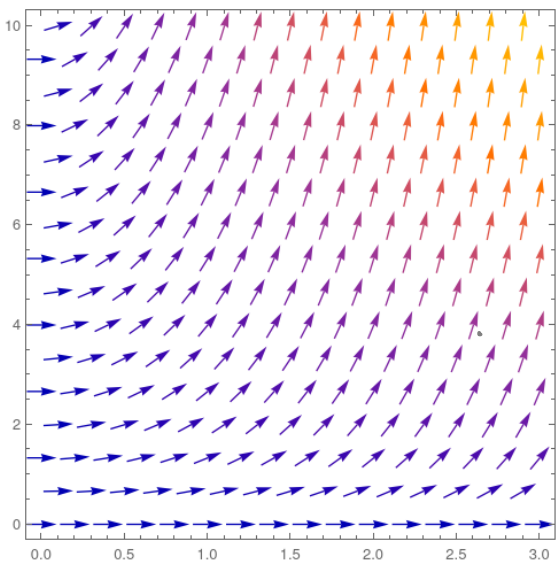
b) $\frac{dy}{dt} = ty$

Draw each of these lines in the direction field.



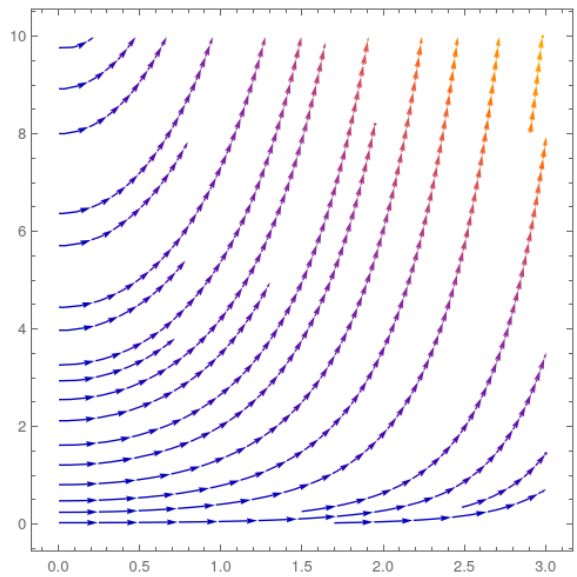
In[18]:= `VectorPlot[{1, (y*t)}, {t, 0, 3}, {y, 0, 10}]`

Out[18]=



In[21]:= `StreamPlot[{1, (y*t)}, {t, 0, 3}, {y, 0, 10}]`

Out[21]=



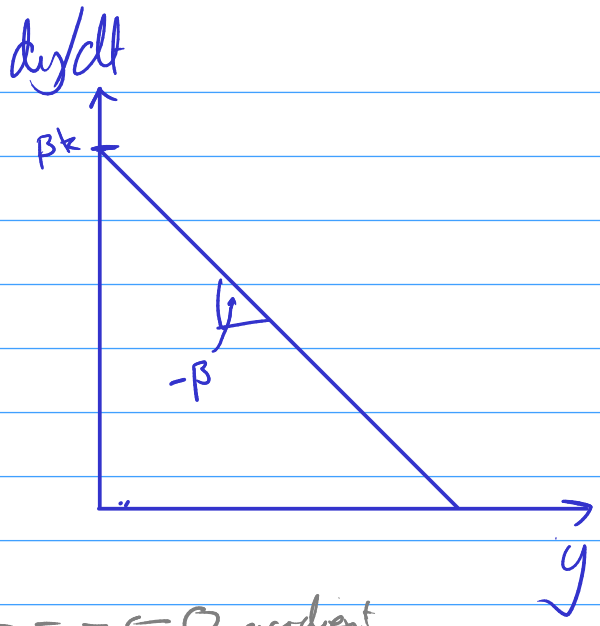
Solving $\frac{dy}{dt} = yt \Rightarrow \int \frac{dy}{y} = \int t\,dt$

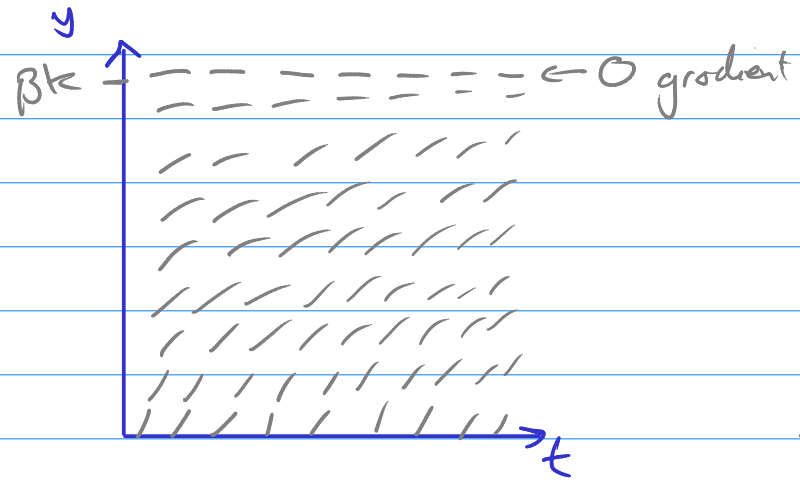
$\ln(y) + c = t^2/2$

$y = e^{t^2/2 - c}$. So we see the effects of changing c in the direction plots

c) $\frac{dy}{dt} = \beta(k-y)$

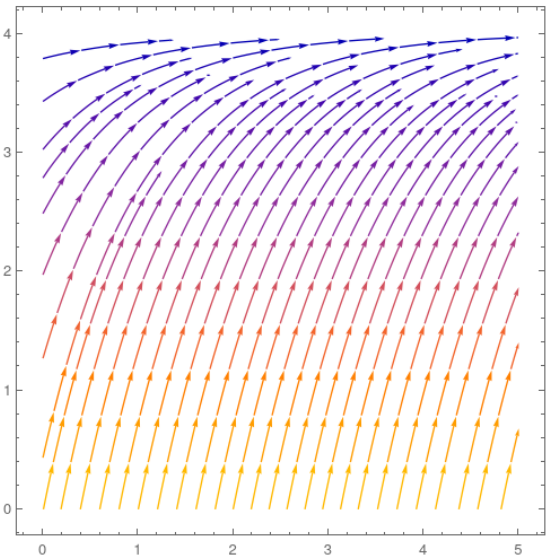
$= \beta k - \beta y$



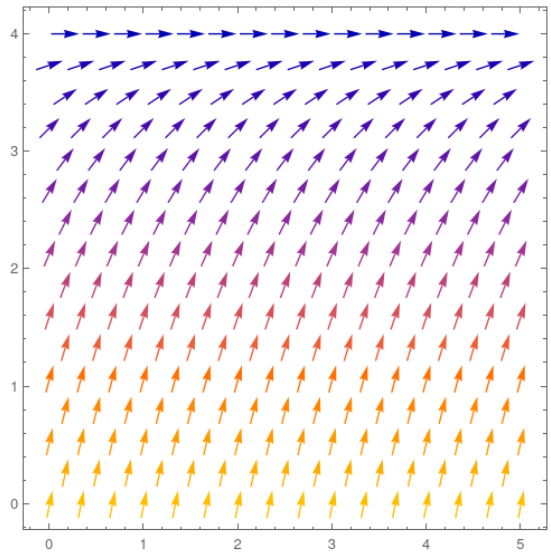


In[34]:= `StreamPlot[{1, ((4 - y))}, {t, 0, 5}, {y, 0, 4}]`

Out[34]=



`VectorPlot[{1, ((4 - y))}, {t, 0, 5}, {y, 0, 4}]`



This is the monomolecular model.

I think you can solve this by now!

Q2) Extension to monomolecular model.

$$\frac{dy}{dt} = \beta e^{-\alpha t}(k-y)$$

β = initial infection rate $(t=0)$

α = infection decay rate. (t^{-1} units)

k = total no of plants.

$$\frac{dy}{(k-y)} = \beta e^{-\alpha t} dt$$

$$-\ln(k-y) = -\frac{\beta}{\alpha} e^{-\alpha t} + c \qquad y(0) = 0$$

$$\therefore -\ln(k) = -\beta/\alpha + c$$

$$c = \beta/\alpha - \ln(k)$$

Plug in c

$$-\ln(k-y) = \beta/\alpha(1-e^{-\alpha t}) - \ln(k)$$

$$\ln\left(\frac{k}{k-y}\right) = \beta/\alpha(1-e^{-\alpha t})$$

Exponentiate

$$\frac{k}{k-y} = \exp\left(\beta/\alpha\left(1-e^{-\alpha t}\right)\right)$$

$$\frac{k}{\exp\left(\beta/\alpha\left(1-e^{-\alpha t}\right)\right)} = k-y$$

$$y = k\left(1 - \frac{1}{\exp\left(\beta/\alpha\left(1-e^{-\alpha t}\right)\right)}\right)$$

$$= k\left(1 - \exp\left(-\beta/\alpha\left(1-e^{-\alpha t}\right)\right)\right)$$

Let $f(t) = \left(-\beta/\alpha\left(1-e^{-\alpha t}\right)\right)$

$$y = k\left(1-e^{f(t)}\right)$$

c) Verify the solution.

Let's differentiate our solution

$$y = k - ke^{f(t)}$$

$$\frac{dy}{dt} = -k\underbrace{f'(t)\cdot\frac{d}{df(t)}e^{f(t)}}_{\text{Chain rule}}$$

$$f'(t) = -\beta e^{-\alpha t}, \quad \frac{d}{df(t)}e^{f(t)} = e^{f(t)}$$

$$\frac{dy}{dt} = \beta k e^{-\alpha t} e^{f(t)} \leftarrow \text{Our solution}$$

$$\frac{dy}{dt} = \beta e^{-\alpha t}(k - y) \leftarrow \text{Original ODE}$$

So we must show that

$$k e^{f(t)} = k - y$$

We know that $y = k - k e^{f(t)}$

$$k e^{f(t)} = \cancel{k} - \cancel{k} + k e^{f(t)}$$

$$k e^{f(t)} = k e^{f(t)} \quad \checkmark$$

d) What is the long term size?

We are asking what is y as $t \to \infty$

Let's look at $f(t)$ first

$$f(t) = -\beta/\alpha(1-e^{-\alpha t})$$

as $t \to \infty$ $f(t) \to -\beta/\alpha$

So $e^{f(t)} \to e^{-\beta/\alpha}$ as $t \to \infty$

So $y = k(1-e^{f(t)}) \to k(1-e^{-\beta/\alpha})$ as $t \to \infty$

e) When is population increasing most rapidly?

Same as asking: When is $\frac{dy}{dt}$ at its maximum? We can find this by finding the roots of d^2y/dt^2.

Recall: $\frac{dy}{dt} = \underbrace{\beta e^{-\alpha t}}_{f(t)} \underbrace{(k-y)}_{g(t)}$

Let's use the product rule

$$\frac{d^2y}{dt^2} = f'g + g'f$$

$$f'(t) = \frac{d}{dt}\beta e^{-\alpha t} = -\alpha\beta e^{-\alpha t}$$

$$g'(t) = \frac{d}{dt}k - y = \cancel{\frac{d}{dt}k}^{\,0} - \frac{d}{dt}y = -\frac{dy}{dt}$$

$$= -\beta e^{-\alpha t}(k-y)$$

$$\therefore \frac{d^2y}{dt^2} = -\alpha\beta(k-y)e^{-\alpha t} + -\beta e^{-\alpha t}(k-y)\beta e^{-\alpha t}$$

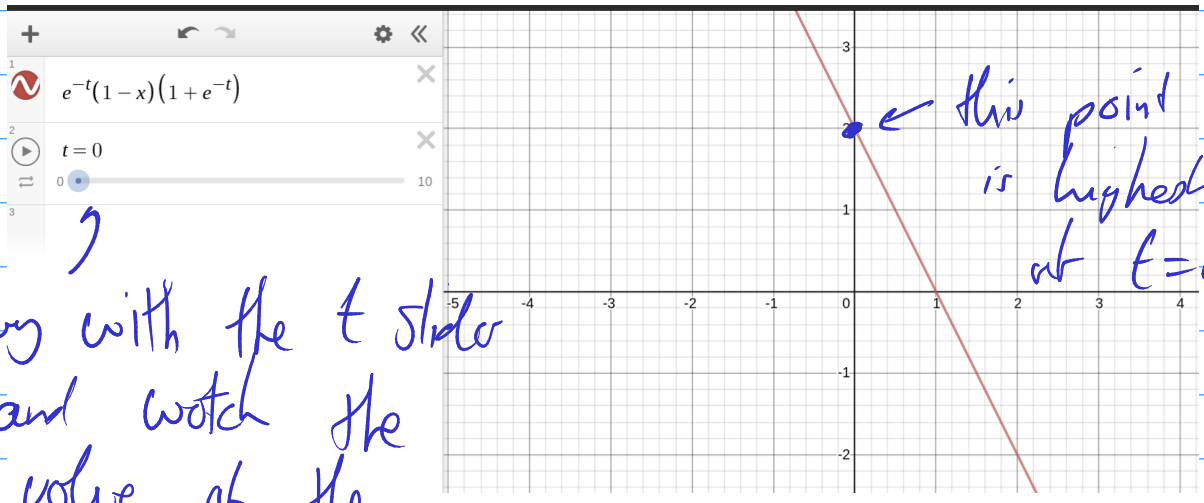
$$= -\alpha\beta(k-y)e^{-\alpha t} + -\beta^2(k-y)e^{-2\alpha t}$$

$$= \beta(k-y)e^{-\alpha t}(-\alpha - \beta e^{-\alpha t})$$

$$= -\beta/\alpha(k-y)e^{-\alpha t}(1 + \beta/\alpha\, e^{-\alpha t})$$

For any given pair of (y,t) we can see that the largest value of y occurs when $t=0$

You can check this by going to desmos and graphing



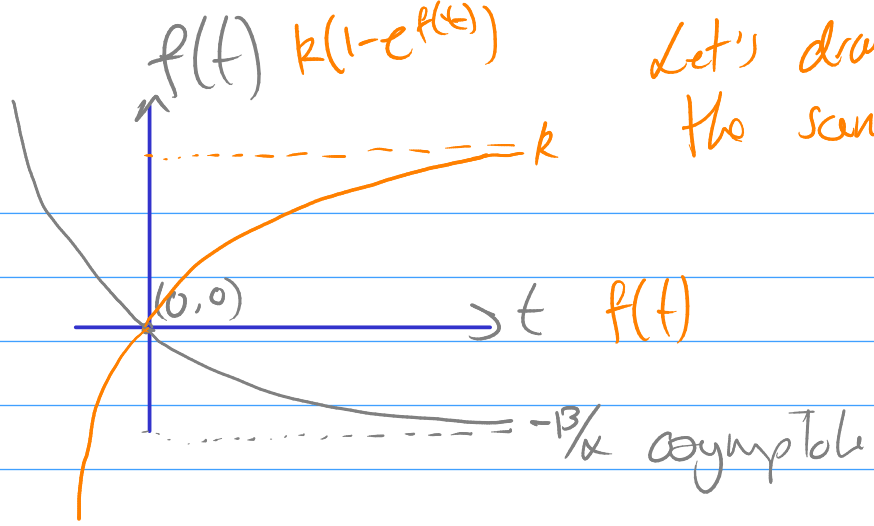
Play with the t slider and watch the value at the intercept change

f) Sketch $y = k(1-e^{f(t)})$

where $f(t) = -\beta/\alpha\,(1-e^{-\alpha t})$

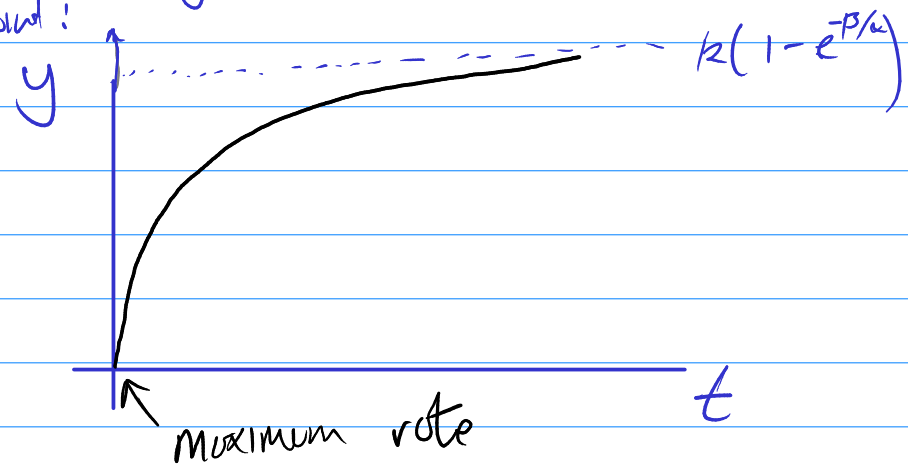
Let's sketch $f(t)$ first:

It's going to have a decay shape because $-\beta/\alpha$ is negative and its intercept is @ 0



Let's draw $k(1-e^{f(t)})$ on the same graph

Now imagine exponentiating $f(t)$ at every single point:



Try using my above examples to generate the direction field using WolframAlpha/Mathematica.

Q3) An extension to logistic growth (again)

$$\frac{dR}{dt} = \beta R(N-R)$$

The rate of rumour spread is proportional to both R (the number who have heard it) and the number yet to hear it $(N-R)$

β is just some rumour spread rate constant.

b) Solving with $R(0) = 0.01$ $(N=1)$

$$\frac{dR}{R(N-R)} = \beta\, dt$$

$$\frac{1}{R(N-R)} = \frac{A}{R} + \frac{B}{N-R}$$

$$\therefore 1 = A(N-R) + BR = AN - AR + BR$$
$$= R(B-A) + AN$$

$\therefore$ $B - A = 1 \Rightarrow B = A$, $AN = 1 \Rightarrow A = 1/N$

$$\frac{1}{R(N-R)} = \frac{1}{RN} + \frac{1}{(N-R)N}$$

$$\therefore \frac{1}{N}\int \frac{1}{R} + \frac{1}{N-R}\, dR = \int \beta\, dt$$

$$= \frac{1}{N}\left(\ln(R) - \ln(N-R)\right) = \beta t + C$$

$$\ln\left(\frac{R}{N-R}\right) = N\beta t + C$$

Let $R(0) = 0.07 = 1\%$

$$\ln\left(\frac{1}{N-1}\right) = \cancel{100\beta \times 0}^{\,0} + C$$

$$C = \ln\left(\frac{1}{99}\right)$$

Particular Solution:

$$R = \frac{N}{1 - 99e^{-\beta N t}}$$

c) Model extension.

$$\frac{dR}{dt} = \beta e^{-\gamma t} R(N-R)$$

$\beta e^{-\gamma t}$ is capturing the lower transmission rate with time. γ is the decay rate.

Solve by separation of variables.

$$\int \frac{dR}{R(N-R)} = \int \beta e^{-\gamma t} \, dt$$

We know the LHS $= \frac{1}{N} \ln\left(\frac{R}{N-R}\right)$

$$RHS = \frac{-\beta}{\gamma} e^{-\gamma t} + C$$

Let $R(0) = 1$ again, $N = 100$

$$\frac{1}{100}\ln\left(\frac{1}{99}\right) = \frac{-\beta}{\gamma} + C$$

$$C = \frac{1}{100}\ln\left(\frac{1}{99}\right) + \beta/\gamma$$

After some algebra:

$$R = \frac{100}{1 + 99\exp\left(-\beta/\gamma \cdot 100(1-e^{-\gamma t})\right)}$$

e) We would like to show that for no more than 75% of students to hear the rumor with $R(0)=1$

$$\gamma > \frac{\beta N}{\ln(297)}$$

Our model must asymptote at 75% or less than

Let's find $R \to \infty$ as $t \to \infty$ in terms of our parameters.

As $t \Rightarrow \infty$ we can see that exponentials go to zero,

$$R(\infty) \rightarrow \frac{100}{1+99\exp\left(-100\beta/\gamma\right)}$$

Let's replace 100 with N for the sake of the question

$$R(\infty) \rightarrow R_{max} = \frac{N}{1+99\exp\left(-100\beta/\gamma\right)}$$

Let us find the combination of N, β, γ which keep this below 75

$$75 > \frac{N}{1+99\exp\left(\frac{-N\beta}{\gamma}\right)}$$ } Solve the inequality for γ

$$75 + 7425\exp\left(\frac{-N\beta}{\gamma}\right) > N$$

$$7425 \exp\left(\frac{-N\beta}{\gamma}\right) > N - 75$$

$$\exp\left(\frac{-N\beta}{\gamma}\right) > \frac{N-75}{7425}$$

$$\frac{-N\beta}{\gamma} > \ln\left((N-75)/7425\right)$$

$$\frac{-N\beta}{\ln\left(\frac{N-75}{7425}\right)} > \gamma$$

$$\frac{-N\beta}{\ln\left(1/297\right)} > \gamma$$

$$\frac{N\beta}{\ln(297)} < \gamma$$

Q4 Von Bertalanffy Model

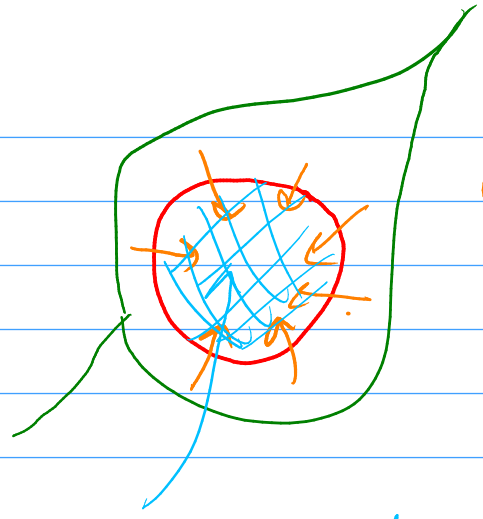
$$\frac{dA}{dt} = \lambda\sqrt{A} - \mu A$$

A = surface area of the leaf.

$$A = \frac{\pi D^2}{4} \qquad \sqrt{A} = \frac{\sqrt{\pi}}{2} D$$

$\lambda \frac{\sqrt{\pi}}{2} D$ is a constant multiple of the circumference (πD)

So the infection growth rate is proportional to circumference, but decreases with surface area



Energy enters through circumference only

Energy consumed over entire surface area.

b) Find equilibrium $\left(dA/dt = 0\right)$

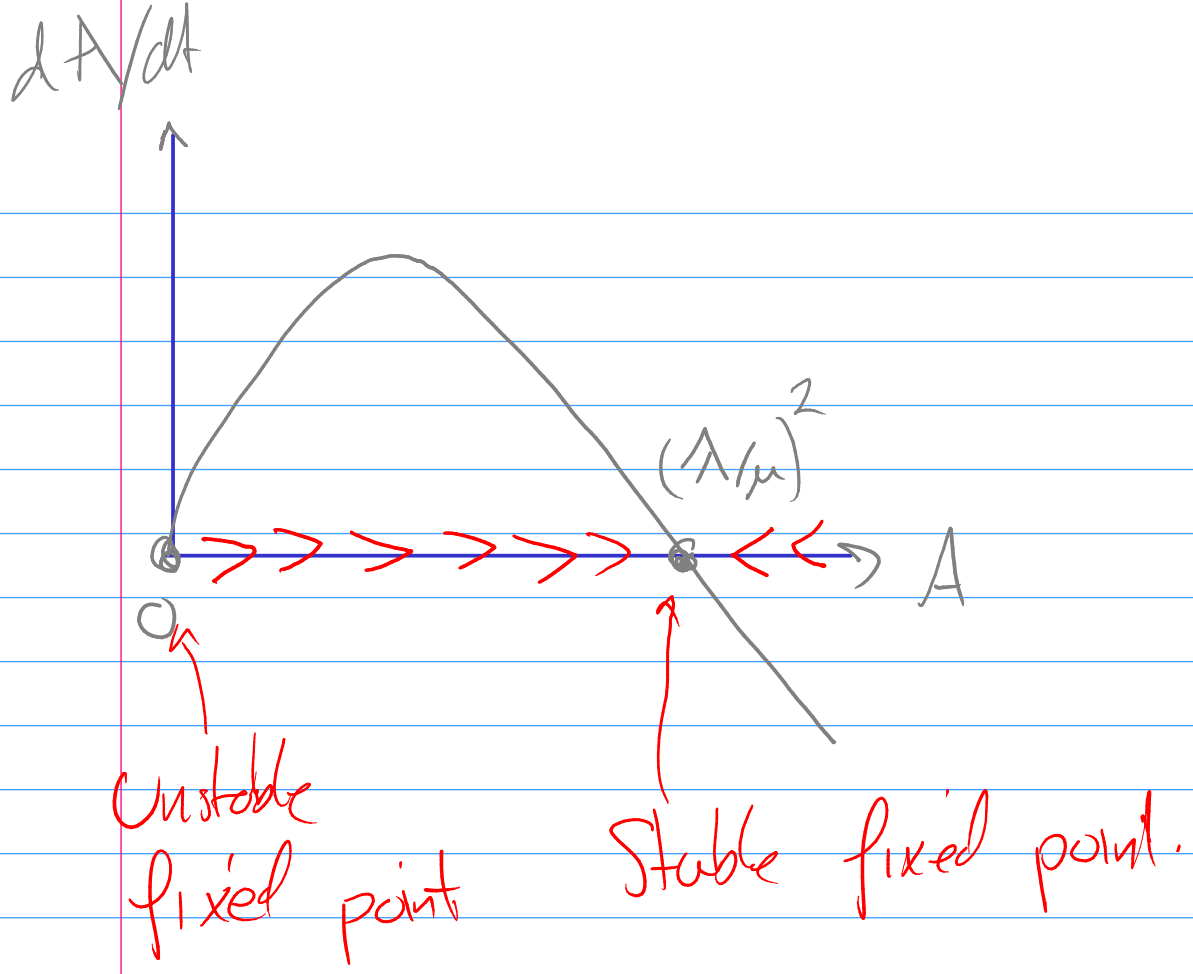
$$0 = \lambda A^{1/2} - \mu A$$

$$= A\left(\lambda A^{-1/2} - \mu\right)$$

$$= \frac{A}{\lambda}\left(A^{-1/2} - \mu/\lambda\right)$$

$\therefore$ Stability of $A = 0$ and $A^{-1/2} = \mu/\lambda$

$$\sqrt{A} = \lambda/\mu \Rightarrow A = \left(\lambda/\mu\right)^2$$



c) Solving with $v = \sqrt{A}$ $(A = v^2)$

Substitute

$$\frac{d}{dt} v^2 = \lambda v - \mu v^2$$

$$= \frac{dv}{dt} \cdot v = \lambda v - \mu v^2 \quad \text{(Cancel } v\text{)}$$

$$\therefore \frac{dv}{dt} = \lambda - \mu v$$

$$\int \frac{dv}{\lambda - \mu v} = \int dt$$

$$\frac{-\ln(\lambda - \mu v)}{\mu} = t + c$$

$$\Rightarrow \frac{-\ln(\lambda - \mu\sqrt{A})}{\mu} = t + c$$

@ $t = 0$, $A = A_0$

$$\frac{\ln(\lambda - \mu\sqrt{A_0})}{\mu} = c$$

After some algebra:

$$A = \left(\frac{\lambda}{\mu}\right)^2 \left(1 - \left(1 - \frac{\mu\sqrt{A_0}}{\lambda}\right) e^{-\mu t/2}\right)^2$$

As $t \to \infty$

$$A = \left(\frac{\lambda}{\mu}\right)^2 \left(1 - \overbrace{\left(1 - \frac{\mu\sqrt{A_0}}{\lambda}\right) e^{-\mu t/2}}^{\text{Goes to } 0}\right)^2$$

And we are left with

$$A_\infty = \left(\frac{\lambda}{\mu}\right)^2$$ ← We found this in part a)

e) When is A increasing fastest?

At maximum $\frac{dA}{dt}$

$$= \frac{d^2A}{dt^2} = 0$$

$$\frac{d^2A}{dt^2} = \frac{d}{dt}\left(\frac{dA}{dt}\right)$$

$$= \frac{d}{dt}\left(\lambda\sqrt{A} - \mu A\right)$$

$$= \frac{\lambda}{2\sqrt{A}} - \mu$$

Set to 0

$$0 = \lambda/2\sqrt{A} - \mu \Rightarrow \lambda/2\sqrt{A} = \mu$$

$$\sqrt{A} = \frac{\lambda}{2\mu} \Rightarrow A = \frac{\lambda^2}{4\mu^2}$$

We must find the corresponding time as well.

Sub into:

$$A = \left(\frac{\lambda}{\mu}\right)^2 \left(1 - \left(1 - \frac{\mu\sqrt{A_0}}{\lambda}\right) e^{-\mu t/2}\right)^2$$

and solve for t

A lot of algebra should yield:

$$t = \frac{2}{\mu} \ln\left(2\left(1 - \frac{\mu\sqrt{A_0}}{\lambda}\right)\right)$$

Q5) Plant growth model.

$$\frac{dN}{dt} = \mu(t) N$$

where $\frac{d\mu}{dt} = -D\mu_0$

$\therefore$ with $\mu(0) = \mu_0$:

$$\int_{\mu_0}^{\mu} d\mu = -\mu_0 D \int_0^t dt$$

$$\mu - \mu_0 = -Dt\mu_0$$

$$\mu = \mu_0 - Dt\mu_0$$

Sub into $\frac{dN}{dt}$

$$\frac{dN}{dt} = (\mu_0 - \mu_0 D t) N$$

Solve

$$\int \frac{dN}{N} = \int \mu_0 - \mu_0 D t \, dt$$

$$\ln(N) = \mu_0 t - \frac{1}{2} D t^2 \mu_0 + C$$

@ $t = 0$, $N = N_0$

$$\ln(N_0) = C$$

$$\ln\left(\frac{N}{N_0}\right) = \mu_0 t - \frac{1}{2} D t^2 \mu_0$$

$$N = N_0 \exp\left(\mu t - \frac{1}{2} D t^2 \mu_0\right)$$

$$= N_0 \exp\left(\mu_0 \left(t - \frac{1}{2} D t^2\right)\right)$$

We want to find when $\frac{dN}{dt}$ is max/min,

So find $\frac{d^2 N}{dt^2} = 0$

$$\frac{d}{dt}\Big(\underbrace{\mu_0(1 - Dt)}_{f}\underbrace{N}_{g}\Big)$$

$$\frac{d^2N}{dt^2} = \mu_0 D N + \frac{dN}{dt}\mu_0(1 - Dt)$$

Substitute dN/dt

$$\frac{d^2N}{dt^2} = \mu_0\left(DN + (1-Dt)\mu_0(1-Dt)N\right)$$